

Fig. 3: Omnipolar Hall effect switch

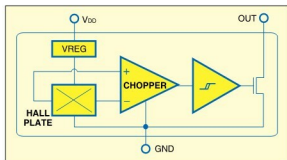


Fig. 4: Unipolar Hall effect switch

rates or can incorporate a magnetic field, a Hall sensor will perform the task.

In a sensing device, the Hall sensor senses the field produced by the magnetic system. The magnetic system responds to the physical quantity to be sensed (temperature, pressure, position, etc) through the input interface. The output interface converts the electrical signal from the Hall sensor into a signal that meets the requirements of the application.

Analogue and digital output

In an analogue (linear) output sensor, the output voltage is proportional to the magnetic field to which it is exposed. The sensed magnetic field

can be either positive or negative. As a result, the output of the amplifier is either positive or negative, thus requiring both plus and minus (dual) power supply.

To avoid the requirement for a dual power supply, a fixed offset (or bias) is introduced into the differential amplifier.

The bias value appears on the output when no magnetic field is present and is referred to as a null voltage. When a positive magnetic field is sensed, the output increases above the null voltage. Conversely, when a negative magnetic field is sensed, the output decreases below the null voltage, but remains positive.

In brief, continuous-time, ratio-metric, linear Hall effect sensors are optimised to accurately provide a voltage output that is proportional to an applied magnetic field. These are well-suited for use in position sensing systems, for both linear target motion and rotational target motion.

Hall effect sensors are sometimes referred to as 'switches' rather than

'sensors' because of the on-off 'digital' voltage signal they output. A digital-output sensor has an output that is just one of two states – 'on' or 'off.'

The basic analogue-output structure can be converted into a digital-output structure with the addition of a Schmitt trigger circuit with hysteresis (see Figs 1 and 2). The Schmitt trigger compares

the output of the differential amplifier with a preset reference. When the amplifier output exceeds the reference, the Schmitt trigger turns on. Conversely, when the output of the amplifier falls below the reference point, the output of the Schmitt trigger turns off.

Usually, digital-output types are widely used as contactless switches in consumer products. One example is the cover switch in clam-shell cell-phones.

Omnipolar and unipolar

Fig. 3 shows the functional block diagram of an omnipolar (north- or south-pole) Hall effect switch. It is based on two Hall effect plates and a chopper-stabilised architecture. The single open-drain output switches on with either a north-pole or south-pole magnetic field of sufficient strength. When the magnetic flux density (B) is larger than operate point (Bop), the output switches on (output pin is pulled low). The output turns off when B becomes lower than the release point (Brp). It remains off when there is no magnetic field.

In case of a unipolar Hall effect switch (refer Fig. 4), the output



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